

Core KPI & Analytics Requirements Sign-Off

Document ID: BRD-KPI-2026-V1

Status: Approved Baseline

Target Release: Product Increment 1

This document defines the core operational Key Performance Indicators (KPIs) and analytical business expressions mandated by leadership. The engineering team must implement these exact definitions within the Phase 1 semantic layer to guarantee data standardization across all digital channels and brick-and-mortar storefronts.

1. Executive Strategic Focus & Business Problem

A cross-functional audit indicates clear top-line revenue strength masked by intense bottom-line compression. The primary objective of this delivery cycle is to map a high-precision business intelligence layer capable of isolating structural margin leakages, evaluating operational return rate intensities, and performing advanced regional and consumer demographic deep dives.

2. Financial Health Engine Requirements

The analytics ecosystem must calculate true transactional yields by translating raw parameters into standardized financial metrics. All evaluations must be structured via explicit definitions to safeguard contextual execution rules.

2.1 Average Order Value (AOV)

Determines the gross spend velocity per transaction across channels to track basket sizes accurately.

BUSINESS LOGIC ENGINE:

$AOV = Total\ Gross\ Revenue / Total\ Order\ Count$

2.2 Profit Margin Percentage (%)

Evaluates true bottom-line retention capability after accounting for Cost of Goods Sold (COGS). It isolates absolute business profitability from top-line sales volume spikes.

BUSINESS LOGIC ENGINE:

$Net\ Profit = Total\ Gross\ Revenue - Total\ COGS$
 $Profit\ Margin\ \% = (Net\ Profit / Total\ Gross\ Revenue) \times 100$

3. Operational Risk & Return Rate Benchmarking

Standardized standalone return calculations fail to expose operational drag when low-volume product lines skew raw percentages. The business requires a relative metrics baseline to track risk intensity safely.

3.1 Relative Return Benchmark Matrix (Return Rate2)

The system must scale the return performance of every individual product tier directly against the company's highest-volume revenue anchor—the **Electronics** division.

BUSINESS LOGIC ENGINE:

Category Return Rate = Total Product Units Returned / Total Product Units Sold

Return Rate2 = Category Return Rate / Electronics Return Rate

3.2 Automated System Exception Rule

To eliminate manual overhead, the semantic reporting layer must deploy a programmatic threshold checker to identify operational profit traps:

- **Rule Trigger Parameters:** If a category contributes **< 1.0%** of corporate sales volume but presents a relative return intensity **> 60.0%** against the Electronics anchor, it must be flagged for immediate decommissioning review.
- **Empirical Anomaly Capture:** The **Grocery Division** breaches this gate (0.37% Sales contribution vs. 61.58% Relative Return Intensity) and must be visually isolated using dynamic alerts.

4. Core Analytical Views Mandated

To deliver a comprehensive BI environment, developers must structure data delivery elements into six distinct interactive views:

Seq	Target Dashboard Sheet	Core KPI Metrics	Primary Analytical Stakeholder
1	AOV & Profit Margin	AOV, Profit Margin %	Chief Financial Officer (CFO)
2	Bottom Line & Top Line Matrix	Gross Sales, COGS, Net Revenue	Corporate Steering Committee
3	Return Rate Analytics	Return Rate, Return Rate2	VP of Supply Chain & Logistics
4	Age Group Analysis	Demographic 10-Yr Cohorts	Director of Performance Marketing

Seq	Target Dashboard Sheet	Core KPI Metrics	Primary Analytical Stakeholder
5	Avg Order Frequency by State	Geographic Order Count Density	Regional Retail Field Managers
6	Electronics Sales Deep-Dive	Sub-Category Margin Split	Inventory Procurement Managers

5. Visual Governance & Slicing Directives

- **Demographic Integrity:** Customer ages must be compiled into fixed, logical 10-year grouping tiers to secure row-context validity against mixed-string format arrays.
- **Universal Data Slicers:** Slicers representing *Geographic State, Region, and Settlement Payment Method* must connect actively to all analytical tables simultaneously to allow seamless cross-filtering.